

1. Which of the following best describes plastic deformation?

- A) Reversible deformation when stress is removed
- B) Irreversible deformation when stress is removed
- C) Deformation under purely elastic conditions
- D) Temporary deformation that returns to zero after unloading

2. Which of the following factors primarily causes cold work hardening?

- A) Reduction in grain size
- B) Formation of dislocations and their interactions
- C) Diffusion of atoms
- D) Increase in temperature

3. Which of the following is a primary effect of cold working on the mechanical properties of metals?

- A) Increase in ductility and reduction in hardness
- B) Increase in yield strength and tensile strength
- C) Decrease in dislocation density
- D) Improvement in grain size

4. What is the main difference between recovery and recrystallization in metals?

- A) Recovery leads to grain growth, while recrystallization does not
- B) Recovery occurs at higher temperatures than recrystallization
- C) Recovery reduces dislocation density, while recrystallization forms new strain-free grains
- D) Recovery increases the dislocation density, while recrystallization decreases it

5. Which of the following describes the "strain aging" phenomenon?

- A) Aging of materials that occurs over time without external force
- B) Increase in strength and hardness due to diffusion of solute atoms to dislocations
- C) Reduction in dislocation density due to annealing
- D) Loss of material strength due to exposure to corrosive environments

6. Which of the following statements about "work hardening" is correct?

- A) It increases the dislocation density, leading to an increase in hardness
- B) It decreases the dislocation density, increasing the yield strength
- C) It decreases the ductility and increases the toughness
- D) It increases the grain size, leading to a stronger material

7. Which material property is most affected during the process of plastic deformation?

- A) Elastic modulus
- B) Yield strength
- C) Density
- D) Poisson's ratio

8. What is the primary mechanism responsible for plastic deformation in crystalline solids?

- A) Grain boundary sliding
- B) Twin formation
- C) Dislocation motion
- D) Vacancy diffusion

9. Which of the following conditions would promote dynamic strain aging?

- A) High strain rate and high temperature
- B) Low strain rate and high temperature
- C) Low strain rate and low temperature
- D) High strain rate and low temperature

10. What is the primary consequence of dynamic recrystallization in metals?

- A) Reduction in dislocation density and formation of new grains
- B) Increase in dislocation density and grain refinement
- C) Increase in strength and decrease in ductility
- D) Formation of vacancies and reduction in atomic diffusion